

GIBSON: AR/VR synchronized city walking system

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Figure 1: GIBSON System: (Left) VR user's view. VR user is walking with AR user in a virtual space while AR user is streaming the cityscape. (Center) VR user uses Oculus Quest 2 to enter the virtual space, (Right) AR user uses Nreal Light to walk with VR user

ABSTRACT

GIBSON is a novel city walking system that enables distant users to walk together as if they are physically in the same city.

The advancement of virtual reality technology has opened the possibility to travel around the world virtually beyond geographical limitations, but there is still room for improvement to make the experience as realistic as real travel. Unlike conventional virtual travel tools and prior multi-user collaborative XR studies, we designed our system to evoke both a sense of co-presence and a sense of being in the real space. For this purpose, we implemented two main functions: (1) function to transfer real-time audio-visual information of the surroundings and (2) function to transfer body movements of users through avatars. We also combined visual positioning system (VPS) and SLAM to align the user locations.

We conducted user testing to verify the experience of cross-AR/VR city walking using GIBSON. The result suggests that our system could make people feel as if they were walking together in the city even though they are physically distanced.

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1 INTRODUCTION

The advancement of virtual reality technology has opened the possibility to travel around the world virtually beyond geographical limitations, but there is still room for improvement to make the experience as realistic as real travel. To pursue more realistic virtual traveling experience for multiple users than conventional VR travel tools, we suppose the following two factors play important role in improving the cross-reality travel experience (a) a sense of co-presence and (b) a sense of being in the real space.

There have been efforts to realize multi-user collaborative XR experience that brings either (a) or (b) above, but not fully brings both (a) and (b). Kasahara et al developed a telexistence system that allows a user in remote location to “Jack-in” the field of view of the local user [Kasahara and Rekimoto, 2015]. However, the system was not designed to let users recognize each other. Singh et al uses GPS and VISLAM and builds a system to synchronize the location information between virtual space and physical space so that users can share their positions and display each other's avatar while the local user walk around a city [Singh et al., 2020]. Although the system shows a novel approach to produce a sense of co-presence between the remote users, the system was not designed to transmit visual information of the surroundings of the local user so that the remote user can see what is taking place in the local side. We propose GIBSON, an AR/VR synchronized city walking system, which evokes both a sense of co-presence and a sense of being in the real space.

2 GIBSON SYSTEM

With GIBSON, a local AR user and a remote VR user log into the system and walk around the city together. AR user uses Nreal Light AR glasses, and remote VR user uses Oculus Quest 2 to log into

the VR space made with MLIT's 3D city model [MLIT, 2020]. To encourage mutual communication between remote users just as they are physically in the same city to walk together, GIBSON system consists of two main functions: (1) function to transfer real-time audio-visual information of the surroundings and (2) function to transfer body movements of users through avatars.

2.1 Transferring real-time audio-visual information of the surroundings

Transferring real-time audio-visual information of the surroundings is key to making people feel that they are sharing the same space and experience. We have implemented functions that transmit the scene of each other's space across AR/VR in real-time: video streaming and voice chat.

We presumed that video streaming would be an effective way to convey the cityscape from the AR user to the VR user since the VR space is pre-made and does not reflect what is taking place in the real space. The AR user can use an RGB camera on the smartphone to stream the video in real time. The video streamed by the AR user plays a role as a showcase window that shares the current cityscape with the VR user. [Figure 1] The video is always displayed in front of AR user so that the VR user can see the real world through the streamed video while walking next to AR user.

Voice chat function transmits audio input from the microphones on Nreal Light and Oculus Quest 2. Not only the voice of the users is transmitted to their counterparts, but also the background sounds from their surroundings, such as voice of pedestrians and sound of cars, are transmitted.

2.2 Transferring user's body movements

For users to feel as if they are together, showing each other's body movements is important to feel the other user's presence.

GIBSON synchronized the positions of the AR/VR avatars so that the users can feel each other's presence while walking around the city. Furthermore, the user's head and body movements were captured and shared to the other side, and the users can feel the presence of the other user. We aligned the vertical position of user avatars so that users can see each other's eyes.

3 LOCATION ALIGNMENT IN GIBSON

To realize the above two main functions, it is necessary to constantly synchronize the positions of the users between the VR space and the real space with high accuracy. We used Visual Positioning System (VPS) and SLAM to align the locations of users between the VR space and the real space.

We used the built-in SLAM of Nreal Light to track the AR user's position and to reflect it to the avatar's position. We also used Visual Positioning System (VPS) for localization of AR user because SLAM is only able to track the relative change in position. We need to acquire the position of the AR user in the real space (latitude and longitude) at the beginning of the experience and when the error in SLAM is accumulated due to continued use.

The reason why we use VPS for localization is because we can expect higher accuracy with VPS than other localization methods. For localization in outdoor AR applications, GPS and vision-based approaches are typically used [Dad et al., 2018]. A study suggested

that a vision-based approach provided higher accuracy than GPS in a pedestrian guiding system in urban areas [Treuillet and Royer, 2010]. Thus, to achieve higher accuracy and better usability, we used Immersal's VPS [Immersal, 2019]. By combining VPS and SLAM, we successfully aligned the locations of users within a few meters of error.

4 USER FEEDBACK

We conducted a questionnaire and an interview with four pairs of users to verify the quality of cross-AR/VR communication delivered by GIBSON. The users' average score to the question, "Did you feel as if you were in the same place as your remote partner?" was 6.1 out of 7. Also, we got the following feedbacks from the interview: "I felt a sense of unity by sharing the video" and "I made eye contact with pedestrians and felt like I was in the same space with them." The result suggests the system has capability to evoke a sense of co-presence and a sense of being in the real space.

5 CONSLUSION

We proposed a cross-AR/VR city walking system GIBSON and conducted a user test in Shibuya. The result of the user test suggests that GIBSON system helps distant users to walk around just as they are physically in the same city.

A possible limitation of GIBSON system is scalability. Although the system uses publicly available city 3D model provided by MLIT for scalability, we need to manually convert the model to build the VR space in the current version of the system. Automation of the file conversion is preferable for nation-wide deployment of the system.

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REFERENCES

- Archita Dad, Bhavna Arora, Nida Parker, and Tejal Rachh, 2018. Augmented Reality: Tracking Methods. International journal of engineering research and technology, 5.
- Immersal Ltd. 2019. Immersal - <https://immersal.com/>
- Shunichi Kasahara and Jun Rekimoto, 2015. JackIn head: immersive visual telepresence system with omnidirectional wearable camera for remote collaboration. In Proceedings of the 21st ACM Symposium on Virtual Reality Software and Technology (VRST '15). Association for Computing Machinery, New York, NY, USA, 217–225.
- Ministry of Land, Infrastructure, Transport and Tourism (MLIT) 2020. PLATEAU <https://www.mlit.go.jp/plateau/>
- Shubham Singh, Zengou Ma, Daniele Giunchi, and Anthony Steed, 2020. "Real-time Collaboration Between Mixed Reality Users in Geo-referenced Virtual Environment"
- Sylvie Treuillet and Eric Royer, 2010. Outdoor/Indoor Vision Based Localization for Blind Pedestrian Navigation Assistance. International Journal of Image and Graphics, World Scientific Publishing, 481-496.